

Executive Summary

Machine Learning–Based Late Delivery Risk Prediction in Global Supply Chain Operations

1. Executive Overview

Reliable logistics and timely delivery are essential for economic growth, international trade, customer service, and supply-chain resilience. This project develops a machine learning–based approach for predicting late-delivery risk using historical logistics and order data. The solution combines exploratory data analysis, feature engineering, machine learning, and explainable AI to support proactive decision-making.

2. Supply-Chain Challenge

Global supply chains involve order processing, transportation, warehousing, carrier operations, and final delivery. Disruptions at any stage can cause delays. Traditional monitoring is often reactive. Predictive analytics can identify potential risks earlier, allowing logistics operators and public authorities to prioritize resources and coordinate preventive actions.

3. Key Findings

Analysis identified differences in late-delivery risk across shipping mode, order region, customer segment, scheduled shipping duration, order quantity, discount rate, sales, and profitability. Six engineered indicators were created: Shipping Pressure, High Quantity Flag, High Discount Flag, Express Shipping Flag, Order Complexity, and Profit Margin. These variables provide additional information for predictive modeling.

4. Economic and Service Impact

Late deliveries can increase transportation and operational costs, customer-service expenses, SLA violations, financial penalties, and customer dissatisfaction. Persistent delays can also reduce the reliability and competitiveness of logistics networks. Better early identification of delivery risk can support more efficient and resilient supply-chain operations.

5. Risk-Prediction Approach

The project uses supervised machine learning to predict Late Delivery Risk. Logistic Regression provides an interpretable baseline, Random Forest captures nonlinear relationships, and XGBoost provides an advanced gradient-boosting approach. Models are evaluated using ROC-AUC, Precision, Recall, F1 Score, and a Confusion Matrix. Predicted probabilities are categorized as Low, Medium, or High risk. SHAP-based explainability can help stakeholders understand important prediction factors.

6. Recommended Government Actions

Develop logistics early-warning systems; prioritize critical transport corridors using delivery-risk patterns; improve secure and standardized data sharing among government and private-sector participants; use analytics to support infrastructure planning; promote AI and digital supply-chain technologies; and establish risk-based monitoring so resources are focused on higher-risk shipments, corridors, and operational areas.

7. Expected Benefits

Predictive logistics analytics can provide earlier risk identification, improved supply-chain visibility, more efficient resource allocation, better transportation planning, reduced operational disruption, improved service reliability, stronger public-private coordination, and better evidence for infrastructure investment decisions. The central benefit is a shift from reactive logistics management toward proactive and predictive decision-making.

8. Policy Implications

Government policy can support data-driven logistics through standardized data frameworks, secure data-sharing mechanisms, digital infrastructure, AI adoption programs, predictive monitoring, public-private partnerships, resilient infrastructure investment, and analytics capability development. Predictive systems should remain decision-support tools, with regular validation, bias monitoring, model updates, privacy protection, cybersecurity, transparency, and responsible AI practices.

9. Conclusion

This project demonstrates how historical logistics data can be transformed into actionable intelligence for supply-chain management. Predictive analytics can help identify potential delivery risks earlier and support timely intervention. For government stakeholders, the approach offers a potential framework for strengthening logistics resilience, improving transportation planning, supporting digital transformation, and promoting reliable supply-chain networks.